BHARATIYA VIDYA BHAVAN'S

SARDAR PATEL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

EXTERNAL ACADEMIC AUDIT

Academic Years: 2024–25 & 2025–26

October 02, 2026

About Department

Program Offered	Branch	Year of Estd.	Intake
B.Tech.	Computer Engineering	1995	60
B.Tech.	Computer Engineering	2021	120
B.Tech.	Computer Engineering	2023	240
M.Tech.	Computer Engineering	2011	18
Ph.D.	Computer Engineering	2012	10
Ph.D.	Computer Engineering	2016	20
Ph.D.	Computer Engineering	2019	30

NBA accredited :

- 2013-14 : accredited for 2 years
- 2016-17 : accredited for AY 2017–18 to 2019–20
- 2021-22 : accredited for AY 2021–22 to 2023–24
- 2024-25 : accredited up to 30 June 2027

"NAAC A Grade" Accreditation awarded: June 28, 2025 (Valid until June 2027)

Faculty list (2025-26)

SN	Name	Designation	SN	Name	Designation
1.	Dr. Sudhir Dhage	Professor	16.	Dr. Anuj Tawari	Assistant Prof.
2.	Dr. Surekha Dholay	Associate Prof.	17.	Ms. Soni Bhambar	Assistant Prof.
3.	Dr. Anant Nimkar	Associate Prof.	18.	Prof. Govind Gaundalkar	Assistant Prof.
4.	Dr. Prasenjit Bhavathankar	Associate Prof.	19.	Ms. Yogita Patil	Assistant Prof.
5.	Prof. Anand Godbole	Associate Prof.	20.	Prof. Azman Qureshi	Assistant Prof.
6.	Dr. Nataasha Raul	Assistant Prof.	21.	Prof. Jotsna Bhagat	Assistant Prof.
7.	Dr. Pramod Bide	Assistant Prof.	22.	Dr. Sanjuktarani Jena	Assistant Prof.
8.	Dr. Kailas Devadkar	Assistant Prof.	23.	Mr. Sudeep Mali	Assistant Prof.
9.	Dr. Disha Sail	Assistant Prof.	24.	Prof. Isha Sawalkar	Assistant Prof.
10.	Dr. Jignesh Sisodia	Assistant Prof.	25.	Mr. Ameya Nile	Assistant Prof.
11.	Prof. Reeta Koshy	Assistant Prof.	26.	Mr. Amey Morye	Assistant Prof.
12.	Prof. Jyoti Ramteke	Assistant Prof.	27.	Prof. Vishwas Badhe	Assistant Prof.
13.	Prof. Sunil Ghane	Assistant Prof.	28.	Ms. Aishwarya Nalawade	Assistant Prof.
14.	Prof. Swapnali Kurhade	Assistant Prof.	29.	Ms. Falak Malik	Assistant Prof.
15.	Prof. Abhijeet Salunke	Assistant Prof.	30.	Mr. Anas Ansari	Assistant Prof.



Technical Assistance:

1. Mr. Rajendra Nalawade
2. Mr. Nandankumar Kinare
3. Mrs. Rupali Bhosale

Supporting Staff:

1. Mr. Mangesh Birje
2. Mr. Vishwanath Kamblekar
3. Mr. Prathamesh Bhonde

Vision & Mission of the Institute

INSTITUTE VISION

"To be a leading technical institution constantly pursuing excellence to develop holistic contributors for the society."

INSTITUTE MISSION

- To provide a conducive academic and research environment that stimulates learning, promotes intellectual growth, and ensures professional development of students, faculty, and staff.
- To collaborate with renowned industry, academic institutions, and research laboratories for academics and research to solve real life problems.
- To leverage the strength of alumni to foster industry-relevant coaching and mentoring of the students.
- To nurture values with a sense of social responsibility and inculcate attitude and skills required for holistic development of the students.

Vision & Mission of Department

DEPARTMENT VISION

"To build a strong teaching and research environment in the domain of Computer Engineering to foster innovative, entrepreneurial and socially responsible contributors."

DEPARTMENT MISSION

- To provide state-of-the-art infrastructure and curriculum to enhance the knowledge, skills and capabilities of students.
- To develop an ecosystem conducive to research and development through industry-academia collaboration.
- To inculcate ethical values, professional discipline, leadership qualities, and social responsiveness among students.
- To foster entrepreneurship and innovative problem solving to address real-world socio-economic challenges.

CRITERIA A

CURRICULUM DESIGN

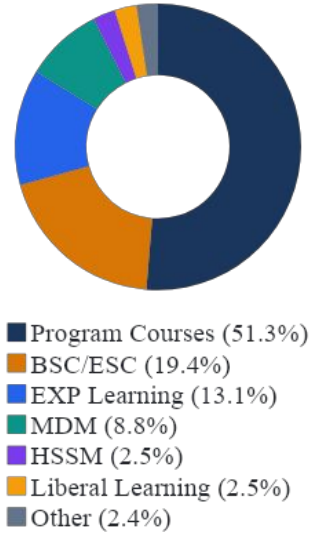
Academic Years: 2024–25 & 2025–26

A.1. The curriculum is well thought and makes sincere attempt to follow the National Guidelines in terms of number of credits of the program, balance of credits for Mathematics, Science, Engineering Science, Program Core, Program Electives and Humanities and Social science related courses.

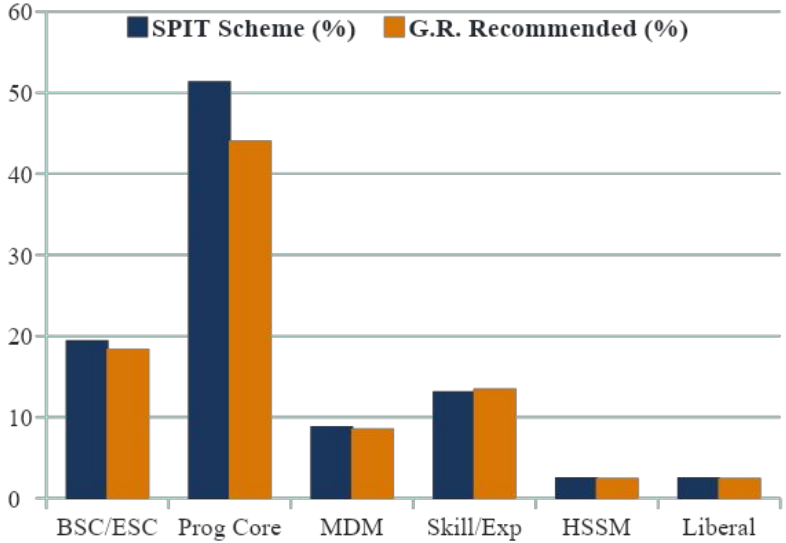
						SPIT					
Sem	BSES	SEC	AEC	HSSM	CC(LLC)	PCC	PEC	OE	EXP LEARNING	MDM	Total
I	11	5	2		1						19
II	11	5	2		1				2		21
III	6	2		2	1	12					23
IV	3	2		2	1	12				3	23
V						17			2	4	23
VI		2				7	6		2	3	20
VII							6	3	4	4	17
VIII								3	11		14
Total	31	16	4	4	4	48	12	6	21	14	160
%	19.38	10	2.5	2.5	2.5	30	7.5	3.75	13.125	8.75	100
G.R. (NEP-2020) Recommended											
Total	30	10	8	4	4	44	20	8	22	14	164
%	18.3	6.1	4.88	2.44	2.44	27	12.2	4.88	13.42	8.54	100

Credit Distribution & Curriculum Structure

Vertical-Wise Allocation of Credits (%)



SPIT Scheme vs. G.R. Recommendation



* The curriculum is well thought and makes sincere attempt to follow the National Guidelines in terms of number of credits of the program, balance of credits for Mathematics, Science, Engineering Science, Program Core, Program Electives, and Humanities.

Link to Curriculum Scheme

Scheme

comp-scheme-2024.pdf

Syllabus

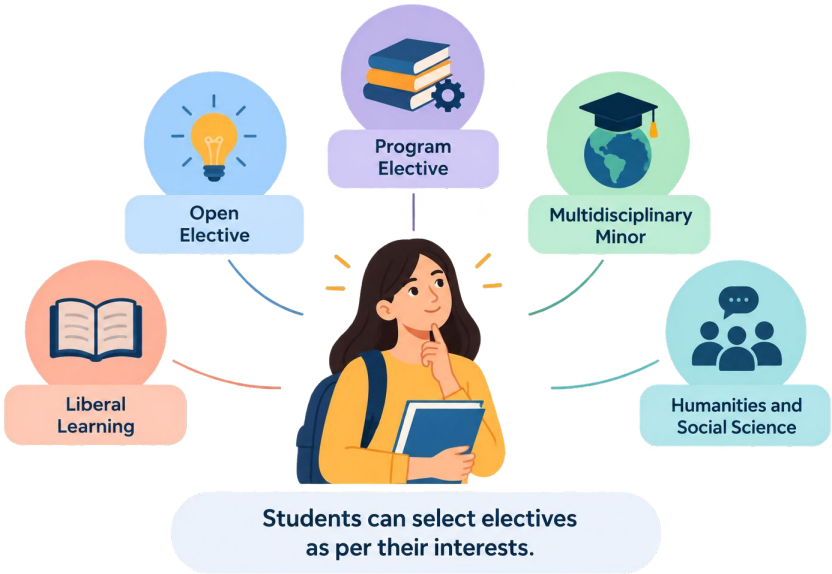
Computer Engineering Syllabus SPIT NEP 2023-2027_v15_Website.docx

A.2. Enough flexibility is incorporated in the curriculum to enable students to take courses of their choice wherever possible. Some sincere efforts towards CBCS are observed.

Flexibility of choice is provided through courses such as,

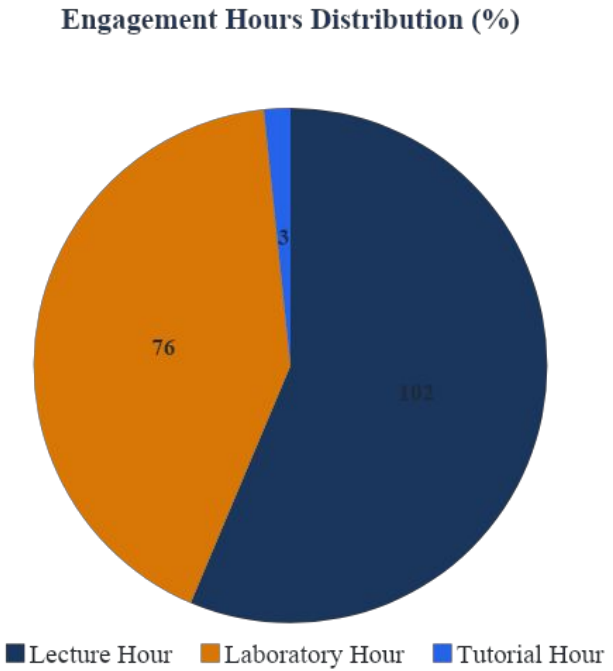
- Program Elective
- Open Elective
- Multidisciplinary Minor
- Liberal Learning
- Humanities and Social Science

Students can select electives as per their interests.

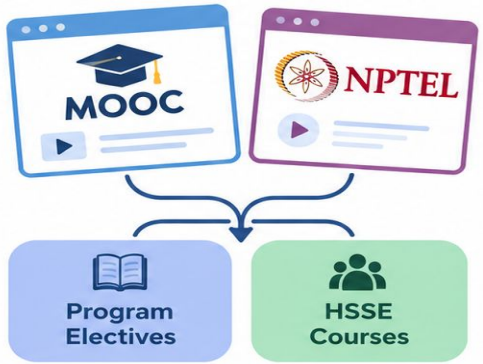


A.3. Curriculum has a balance of theory and laboratory sessions, to ensure strong hands-on learning.

L	Lecture Hour	102
T	Tutorial Hour	3
L	Laboratory Hour	76
O	Other Work (Self Study)	231
T	Total Engagement in Hours	410
C	Credit Assigned	160



A.4. The curriculum encourages the learning from the MOOCs/NPTEL







- MOOC and NPTEL courses are introduced as a part of Program Electives and HSSE courses.
- MOOC courses are selected considering the requirements from the industry and intrack with the syllabus

A.5. Curriculum justifies the Program Outcomes and PSOs articulated Each course syllabus individually maps its course outcomes with PO and PSO.

Pre-requisite Course Codes, if any.	
Course Objective: Apply various algorithmic design paradigms, analyze their asymptotic performance, and synthesize efficient algorithms for common engineering design scenarios.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
CE207.1	Analyze the complexity of algorithms and apply divide-and-conquer strategy to solve problems.
CE207.2	Make use of string matching algorithms to solve problems .
CE207.3	Apply the concept of greedy method, dynamic programming and Maximum Flow.
CE207.4	Apply the idea of backtracking, branch and bound strategy to solve problems.
CE207.5	Make use of NP-completeness and Approximation Algorithms

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CE207.1	3	3	2	1	-	-	1	-	-	-	-	-
CE207.2	3	2	2	1	-	-	1	-	-	-	-	-
CE207.3	3	3	2	1	-	-	1	-	-	-	-	-
CE207.4	3	3	2	1	-	-	1	-	-	-	-	-
CE207.5	2	2	3	1	-	-	1	-	-	-	-	-

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CE207.1	-	-	-	-	-	-	-
CE207.2	-	-	-	-	-	-	-
CE207.3	-	-	-	-	-	-	-
CE207.4	-	-	-	-	-	-	-
CE207.5	-	-	-	-	-	-	-

A.6. Curriculum is well thought, innovative, suits the outcome-based education and has sufficient bandwidth as well as depth of domain.



The curriculum satisfy all the requirements of NEP.

It is designed on the basis of outcome based education framework.

It includes novel ideas such as research internship, MDM, Industry internship and research with honours.

CRITERIA B
TEACHING-LEARNING PROCESS

Instruction methods, collaborative learning, continuous evaluation

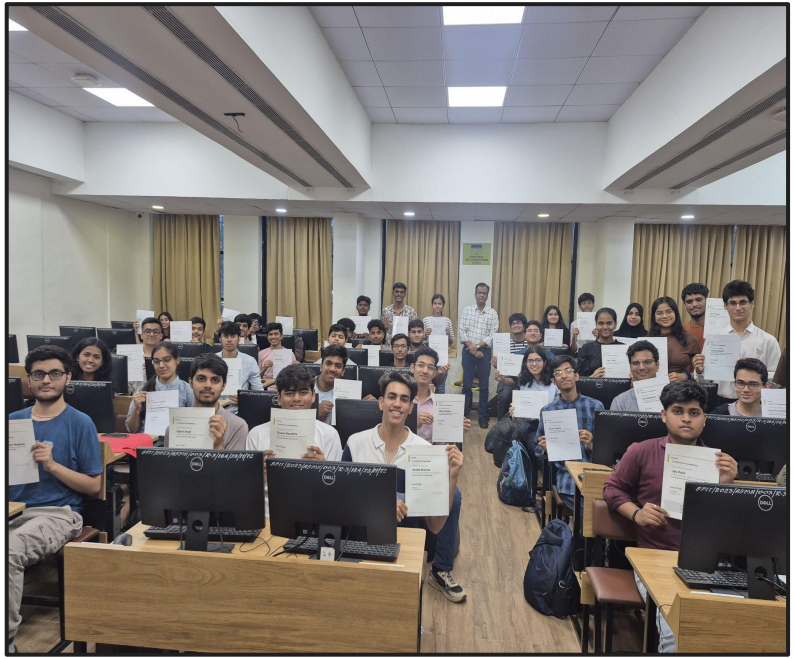
This includes various process such as

- Process to improve instruction methods
- Process for collaborative learning
- Process for improving quality of laboratory experience
- Process for encouraging bright students
- Process for assisting weak students
- Initiatives to improve quality of internal assessment
- Process to improve quality of Mid/End Semester Question Papers
- Process to improve quality of assignments
- Process to improve the evaluation

B.1. Department makes sincere attempts to collaborate for the design and delivery of the teaching and learning processes of the program, tries to follow best practices of higher institutions, and optimizes available resources, in pursuit of the excellence.

- Students are provided digital content for the subjects on moodle LMS.
- Evaluation of students is done using elaborate rubrics that are used in the laboratory for a thorough evaluation..
- Industry provided skill enhancement programs for students from NVIDIA DLI for AI/ML domain.
- Cross domain Skill enhancement programs for students during summer break under the aegis of 'Beyond the Degree'.
- MoU with Linux Professional Institute , Canada to provide curriculum for Cloud, Linux administration, DevOps and also preparing students for industry certifications.
- 24hr. hackathon like AIRAVAT is organised for the students.
- Events like Quantum Computing Conclave for sunrise sectors like Quantum Computing for students and faculty alike.
- FDP's covering latest technologies like GENLANG 5.0 gave the faculty a new avenue to explore and learn.
- Faculty of CE department are encouraged to participate in cross domain STTP like the one on NI LABVIEW delivered by industry partner NI-Emerson.

Cohort of NVIDIA DL program



B.2.Coursework (assignments, quizzes, lesson plan for various courses, laboratory expts etc) offer sufficient breadth and depth appropriate for the degree offered.

- Lesson plans are meticulously designed and mapped to Course Outcomes (COs) to ensure comprehensive topic coverage.
- Integration of real-world case studies to assess practical understanding and application.
- Assignments and quizzes are calibrated across various Bloom's Taxonomy levels to test higher-order thinking.
- Extensive use of Project-Based Learning (PBL) to bridge theoretical concepts with hands-on development.

B.3. Department looks forward for Industry / alumni participation wherever possible, which is demonstrated by few examples of co-teaching, joint project work

Alumni Internship Evaluation

Alumni are involved in evaluation of internship students undergoing internships at various industries. In 2024-25, 18 alumni evaluated the students; in 2025-26, 11 alumni participated in evaluation.

Alumni Talk

Dr. Sudhir Dhage arranged the alumni talk “Exploring and Preparing for management and consulting careers”, delivered by Soumyadeb Bhattacharjee, Strategy and Management Consultant.

Co Teaching

Subject AIML, arranged by Prof. Swapnali Kurhade Delivered by Dr. Maheshkumar H. Kolekar, IIT Patna on 13/10/2024





B.4. Project work is relevant, demonstrates the value addition and is of expected quality from bright undergraduate students.

- **Relevance:** Projects address real-world problems and have clear objectives.
- **Technical Skills:** Students apply appropriate technologies and concepts to develop and test the solution.
- **Innovation & Value Addition:** Projects show new ideas, practical solutions, and meaningful value addition.
- **Research & Publication:** Students are encouraged to publish research papers or file patents based on their completed project work.

B.4. Project work is relevant, demonstrates the value addition and is of expected quality from bright undergraduate students.

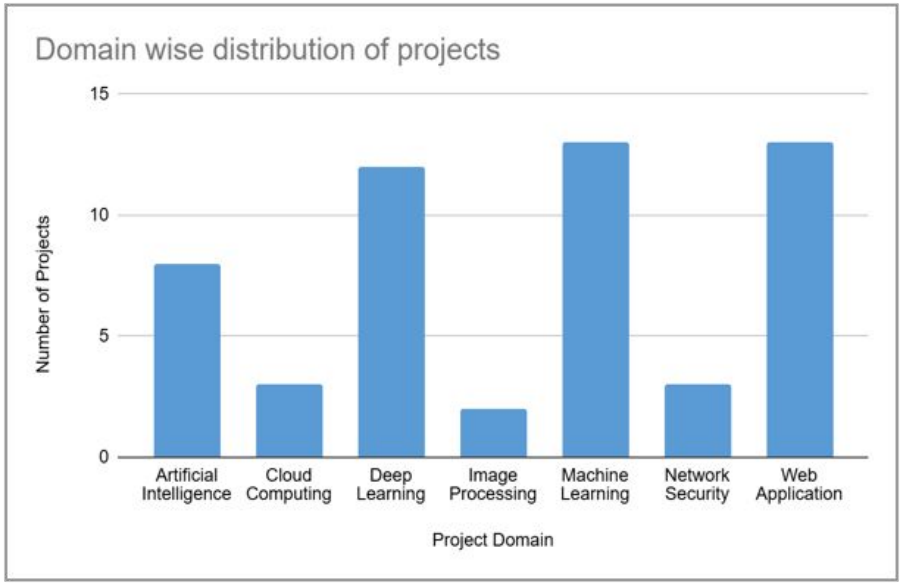


Figure 1: Domain wise percentage of projects 2024-25

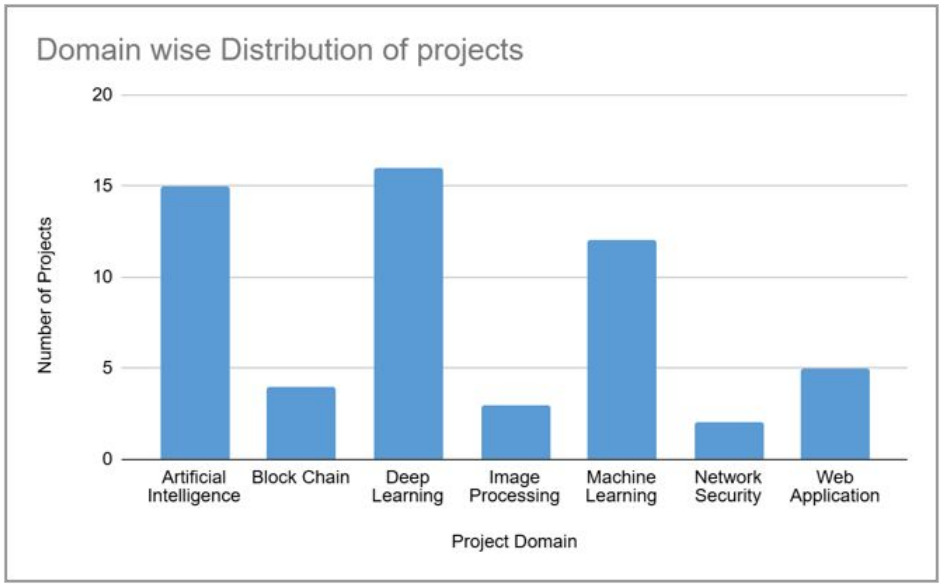


Figure 2: Domain wise percentage of Projects 2025-26

B.5. Department continuously upgrades laboratory infrastructure and year on year investment in the laboratory is sufficient.

Sr. No	Configuration	Quantity	Cost (Rs)	Room No/ Lab
1	HPPRODESK 2 GLA, Desktop-D2YL7AT R5, 16 GB RAM,512 GB SSD, HP 21.5" LED Monitor	20	11,56,400/-	Lab 5 (Room No. 606)
2	HPPRODESK 2 GLA, Desktop-D2YL7AT, R5, 16 GB RAM, 512 GB SSD, HP 21.5" LED Monitor	20	11,56,400/-	Lab 6 (Room No. 606)
3	HPPRODESK 2Gla, Desktop-D2YL7AT, Core I5, 16 GB RAM, 512 GB SSD, HP 21.5" LED Monitor	20	13,15,794	Lab 4 (Room No. 604)
4	HPPRODESK 2Gla, Desktop-D2YL7AT, Core I5, 16 GB RAM, 512 GB SSD, HP 21.5" LED Monitor	20	13,15,794	Lab 8 (Room No. 608)

B.6. Support for slow and physically /socially /financially challenged students, and it is demonstrated by the sufficient evidence.

- **Extra attention to the physically challenged**: Faculty use user-friendly platforms to share notes,lessons. Special efforts to read out questions during exams. Lab sessions conducted with care and empathy to avoid accidents and injuries. Slow learners supported through regular practice assignments.
- **Extra time in exam and extra problem-solving practice** : 25% extra time given to students with learning disability. Slow learners supported through regular practice assignments.
- **Physical facility :** Ramps for access to main building and to other important areas. Lifts with Braille symbols for floor numbers, Disabled friendly washrooms. Wheelchair available at the entrance of the lift and accompanied by housekeeping staff for the convenience of the students.

B.7. Department takes cautious efforts to challenge the bright students and provides them enough opportunities to demonstrate distinctive performance.

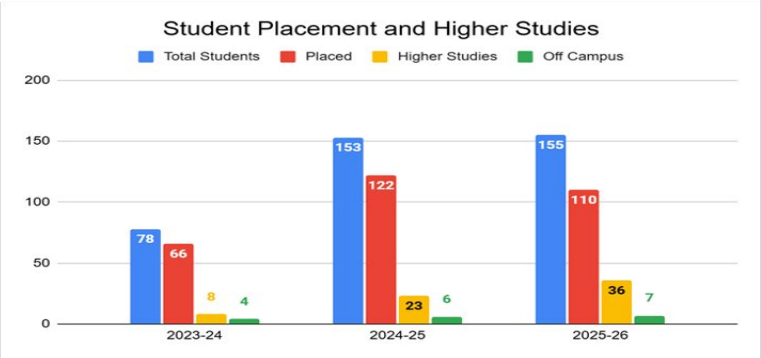
- Identification of Fast learner based on the interest of students through a continuous evaluation process.
- Students encouraged to undertake advanced learning activities beyond the regular curriculum.
- **Example:** implement comprehensive application that integrates simulations for all laboratory experiments.
- Students are encouraged to participate in intercollegiate events, national level contests and pursue research internships.
- Students are motivated to complete certification.
- Industry based resource persons for special sessions.

B.8.Student competency in terms of placements/ success in competitive examinations / selection for higher studies in reputed Universities/ institutes

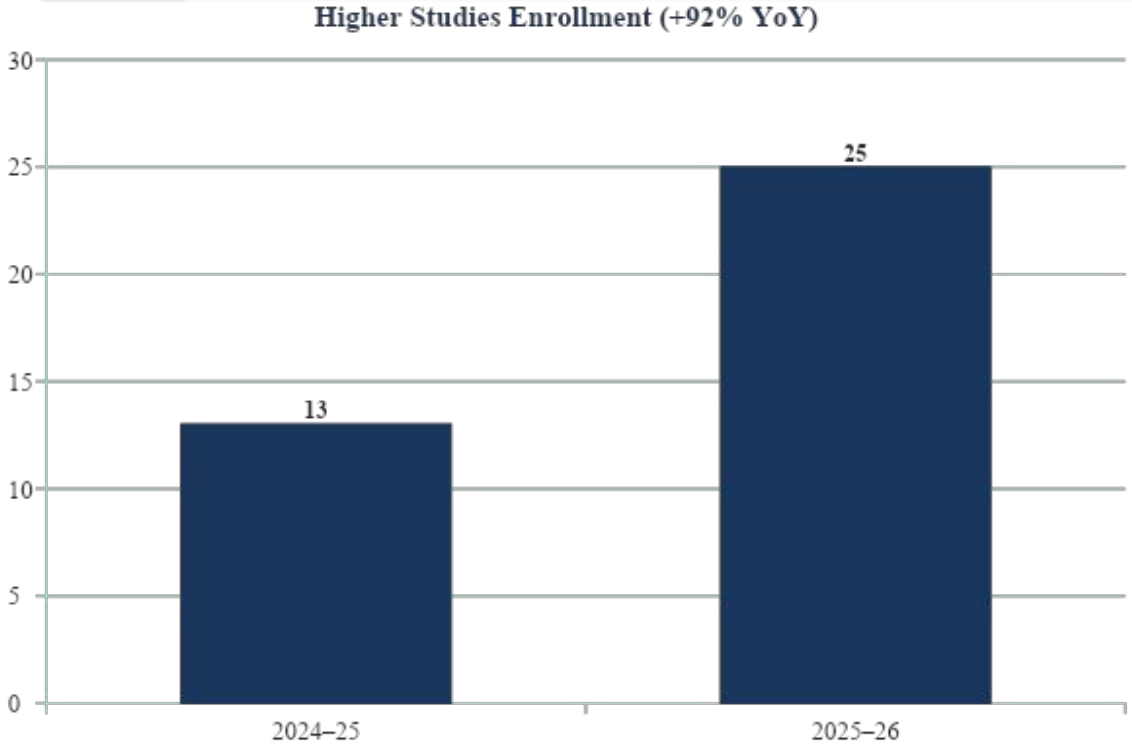
Placement Summary AY - 2025 - 26			
Company Details		Student Details	
Elite	21	Total Students	155
Super Dream	70	Opted for Placements	119
Dream	10	Placed	110
Normal	9	Ineligible	7
Ineligible	7	Placement Percentage	98.21%
Unplaced	2		

Placement Summary - AY 2024 - 25			
Company Details		Student Details	
Elite	47	Total Students	153
Super Dream	55	Opted for Placements	130
Dream	14	Placed	122
Normal	6	Ineligible	5
Ineligible	5	Placement Percentage	97.60%
Unplaced	3	Entrepreneur	0

Placement Summary - AY 2023 -24			
Company Details		Student Details	
Elite	33	Total Students	78
Super Dream	30	Opted for Placements	70
Dream	2	Placed	66
Normal	1	Ineligible	4
Ineligible	4	Placement Percentage	100.00%
Unplaced	0		



Higher Studies 2024-25 and 2025-26





AY	Count
2024-25	13
2025-26	25

CRITERIA C

ASSESSMENT AND EVALUATION

Direct & indirect attainment, continuous evaluation, transparent grading

C.1 Quality of assignments/ test papers/ End Semester Examination papers appears to suit the quality of the students getting admitted

- Question Paper is set as per the blooms taxonomy.
- While setting the question papers , care is taken ,to ensure that all COs are covered including the self study topic.
- For ESE paper 2 sets are drawn for every subject.
- Difficulty Level of the Question Paper is set so as to ensure Gaussian curve.
- Question paper moderation is done by internal faculty other than paper setter. Depending upon the suggestions received concerned faculty update the paper and then it is submitted to examination section.
- Question Paper Audit was carried out by external faculty.

C2.Design and conduct of continuous assessment of theory and lab assignments is deeply thought of

Various assessments methods such as Case Study, Quiz, Viva, Projects, Open Book Test, Poster Presentation, e.t.c. are designed for continuous evaluation for Theory and Lab courses to improve the quality of assessment

Link to Lesson Plan: https://drive.google.com/drive/folders/1kN8Dc457fqZv0e_R12Hv2X38NuQKinjM



C3.Wherever possible industry persons, alumni, external experts are invited for evaluation and assessment.

- Industry professionals, alumni, and external experts are invited, wherever possible, to evaluate projects, internships, and seminars and provide students with practical feedback and guidance.
- Industry internships and research internships at reputed institutes such as IITs, ISRO, NTRO, etc. are evaluated with inputs from our alumni working in relevant industries, ensuring practical and industry-oriented assessment.

Industry Internship Evaluation

	Name of the Faculty	Department/Company	Email ID
Chairman(Internal Supervisor)	Prof. Nataasha Raul	Comp	nataasharaul@spit.ac.in
Member1(Internal Examiner)	Prof. Abhijeet Salunke	Comp	abhijeet_salunke@spit.ac.in
Member2(Industry Supervisor/Alumni)	Deep Mehta (8369650711)	Senior Software Engineer,Morgan Stanley	deep.mehta174@gmail.com

C4. Coverage of syllabus in examinations including ISE, MSE, ESE is fair, and MSE/ESE are designed to have natural gaussian. June 2025

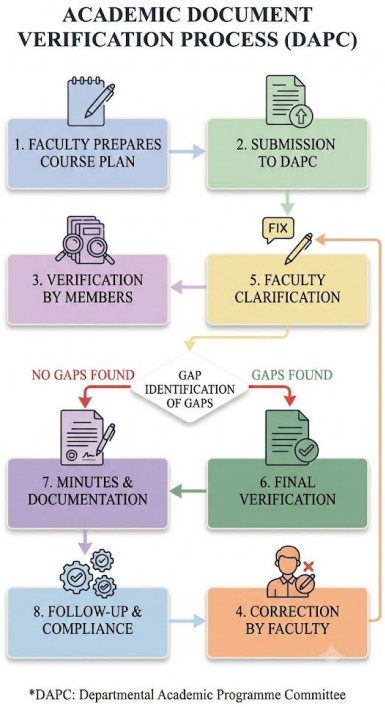
- Faculty Completes the syllabus by conducting extra lectures required if any also conducts remedial lectures if any.
- Weightage for ISE - 20% , MSE is 20% and ESE is 60%.
- upto MSE syllabus coverage is upto 40 to 50% and for ESE it is 100%.
- After 2-3 weeks DAPC meeting gets conducted in which lesson plan is gets approved

Link for SA values

PG even sem 2025 26 SA values.pdf
PG Odd sem 2025 26

DAPC MOM-EVEN SEM-Academic year 2025-2026
DAPC MOM ODD SEM Academic year 2025-26

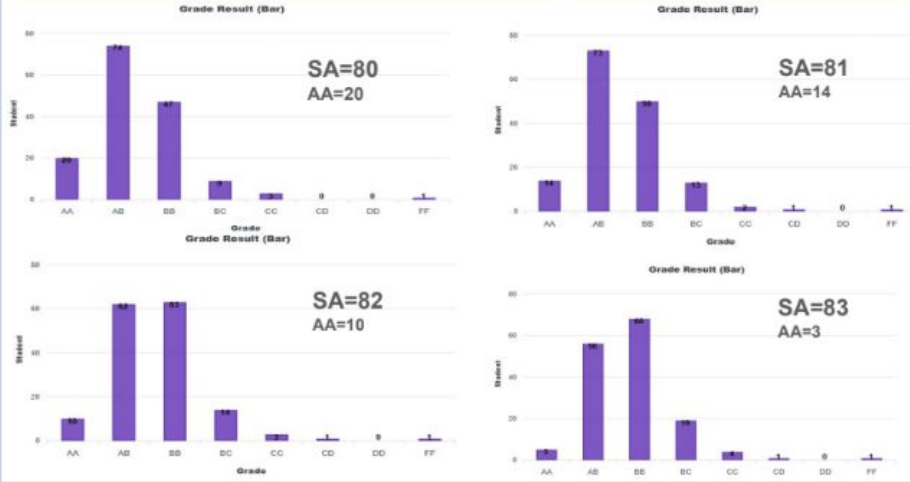
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UG odd sem 2025 26



C4. Coverage of syllabus in examinations including ISE, MSE, ESE is fair, and MSE/ESE are designed to have natural gaussian.

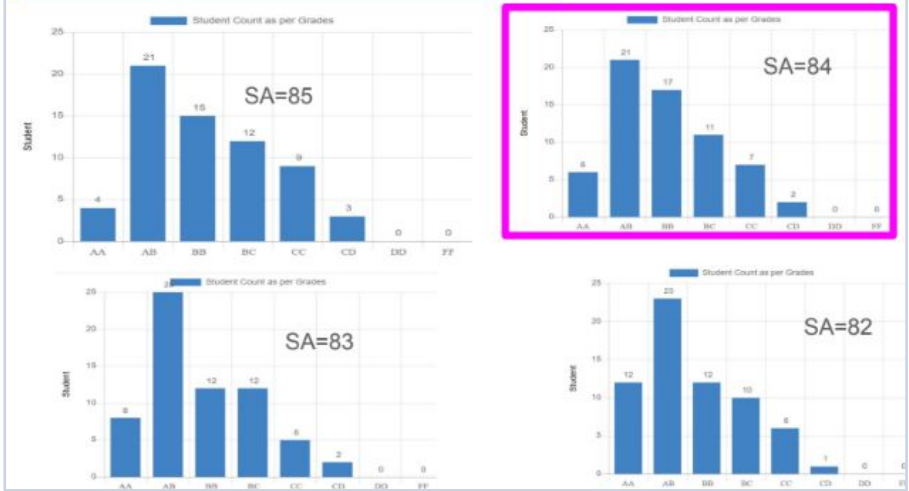
SAMPLE Gaussian Distribution Graph for PE subjects

Course Name - Information Visualization Course Coordinator -Yogita Patil	Course Code - CE332
Semester – VI, **Credit code: 201**	Course Category: PE
No of students enrolled in the course - 154	**Final SA Value - 80**



SAMPLE Gaussian Distribution Graph for OE subjects

Course Name - Software Testing Course Coordinator - Aparna Halbe	Course Code - OEDS3
Semester – VII, **Credit code: 201**	Course Category - OE
No Of students enrolled in the course - 64	**Final SA Value - 84**



C5. Transparency in evaluation including display of marks/ grades secured in every course.

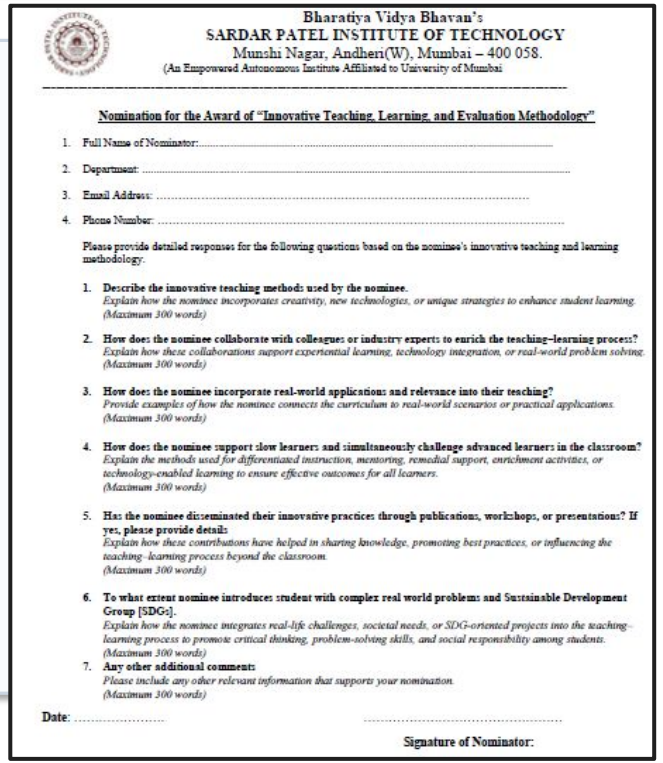
- The ISE ,MSE , ESE dates and grading method along with Rubrics are announced at the start of the course through common platform like Google Classroom ,Email, Moodle ,e.t.c. to maintain transparency in evaluation system.Also , the marks are displayed immediately after the assessment.
- Open house gets conducted for MSE/ESE
- Grievances gets addressed in the open house

C6 Department encourages innovations in the assessment. Identifies the best performing faculty in this aspect.

The institution follows a systematic and transparent process for identifying and recognizing the Best Performing Teacher in Innovative Teaching-Learning and Evaluation (TLE) Practices. The process is coordinated by the Internal Quality Assurance Cell (IQAC) and involves participation from faculty members across all departments.

1. Initiation of the Process by IQAC
2. Submission of Data by Faculty Members
3. Evaluation by the IQAC Team
4. Identification and Recognition of the Best Performing Faculty

View Nomination Form



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Munshi Nagar, Andheri(W), Mumbai – 400 058.
(An Empowered Autonomous Institute Affiliated to University of Mumbai

Nomination for the Award of "Innovative Teaching, Learning, and Evaluation Methodology"

1. Full Name of Nominator: .....
2. Department: .....
3. Email Address: .....
4. Phone Number: .....

Please provide detailed responses for the following questions based on the nominee's innovative teaching and learning methodology.

1. **Describe the innovative teaching methods used by the nominee.**
   Explain how the nominee incorporates creativity, new technologies, or unique strategies to enhance student learning.
   (Maximum 300 words)
2. **How does the nominee collaborate with colleagues or industry experts to enrich the teaching–learning process?**
   Explain how these collaborations support experiential learning, technology integration, or real-world problem solving.
   (Maximum 300 words)
3. **How does the nominee incorporate real-world applications and relevance into their teaching?**
   Provide examples of how the nominee connects the curriculum to real-world scenarios or practical applications.
   (Maximum 300 words)
4. **How does the nominee support slow learners and simultaneously challenge advanced learners in the classroom?**
   Explain the methods used for differentiated instruction, mentoring, remedial support, enrichment activities, or technology-enabled learning to ensure effective outcomes for all learners.
   (Maximum 300 words)
5. **Has the nominee disseminated their innovative practices through publications, workshops, or presentations? If yes, please provide details:**
   Explain how these contributions have helped in sharing knowledge, promoting best practices, or influencing the teaching–learning process beyond the classroom.
   (Maximum 300 words)
6. **To what extent nominee introduces student with complex real world problems and Sustainable Development Group [SDGs].**
   Explain how the nominee integrates real-life challenges, societal needs, or SDG-oriented projects into the teaching–learning process to promote critical thinking, problem-solving skills, and social responsibility among students.
   (Maximum 300 words)
7. **Any other additional comments:**
   Please include any other relevant information that supports your nomination.
   (Maximum 300 words)

Date: .....

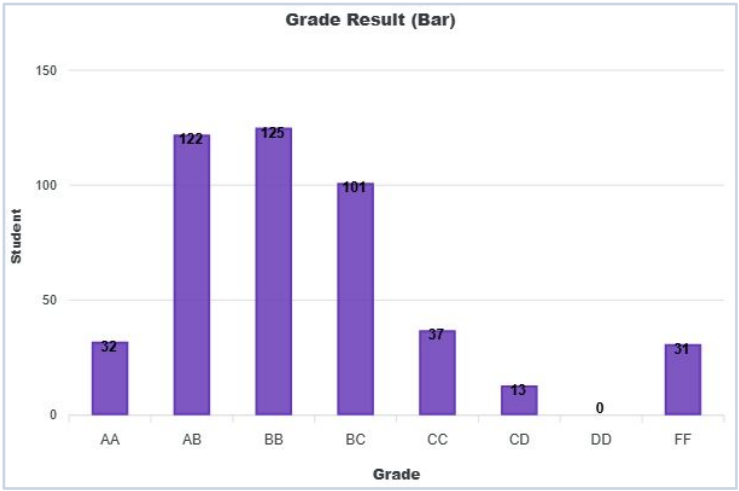
.....

Signature of Nominator:

C7. Department analyzes the results critically, reason out the abnormalities, deliberates in meetings, and finds out the solutions. Extremities are identified and mentored.

- Subject faculty prepares slides with 4 SA value graphs
- Department meeting is convened to analyze SA value
- Final SA value is selected such that %students in AA grade is 10-15% of total students

Course Name - Operating System
No of students enrolled in the course - 451
Final SA Value - 73



Evaluation & Examination Guidelines

- Usually SA Values are used for relative Grading and the result should follow the Gaussian Distribution
- The OPEN house is conducted for MSE and ESE where students can see his/her paper and marks awarded
- Any grievances raised are solved by rechecking of the questions
- If students who are unable to clear the examination, re-exam is held after the results are declared

C8. Quality of assessment tools and extent of attainment of learning outcomes

Theory

- ESE (Weight: 60%, 100 Marks)
  - Written Exam
- MSE (Weight: 20%, 30 Marks)
  - Written Exam
- ISE(Weight: 20%, Marks decided by instructor)
  - Quiz
  - Assignment
  - Open Book test
  - Tutorials
  - Mode approved by DAPC

Lab

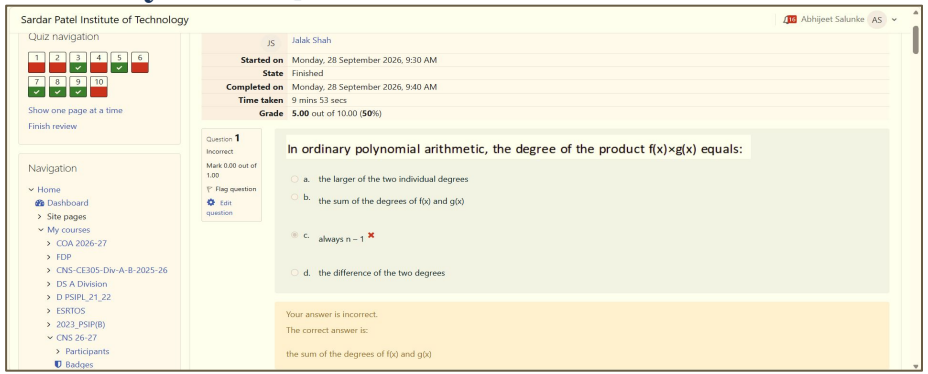
- ISE (Weight: 80%)
  - Experiments evaluation
- ESE(Weight: 20%)
  - Lab exam

C8. Quality of assessment tools and extent of attainment of learning outcomes

Theory-MSE/ESE

Q. No.	Question	Max. Marks	CO-BL
Q. 1	Elon Musk is trying to build a solar city in India. He has purchased a certain number of solar panels. He wants to arrange them into rows such that each row contains the same number of panels. In his first attempt he tried to fit 17 panels in each row, but he had 3 solar panels left in last row. So, then he tried with 13 panels in each row, but he was left with 5 panels in last row. Then he tried 11 panels per row and was left with 7 panels in last row. Based on this information, determine the total number of solar panels Elon Musk has. Show all steps clearly. **OR** The Euclidean algorithm has been known for over 2000 years and has always been a favorite among number theorists. After these many years, there is now a potential competitor, invented by J. Stein in 1961. Stein's algorithm is as follows: Determine gcd(A, B) with A, B ≥ 1. ***STEP 1*** *Set* $A_1 = A$, $B_1 = B$, $C_1 = 1$ ***STEP 2*** *For* $n > 1$, *(1) If* $A_n = B_n$, *stop.* $gcd(A, B) = A_n C_n$ *(2) If* A_n *and* B_n *are both even, set* $A_{n+1} = A_n/2$, $B_{n+1} = B_n/2$, $C_{n+1} = 2C_n$ *(3) If* A_n *is even and* B_n *is odd, set* $A_{n+1} = A_n/2$, $B_{n+1} = B_n$, $C_{n+1} = C_n$ *(4) If* A_n *is odd and* B_n *is even, set* $A_{n+1} = A_n$, $B_{n+1} = B_n/2$, $C_{n+1} = C_n$ *(5) If* A_n *and* B_n *are both odd, set* $A_{n+1} = \lvert A_n - B_n \rvert$, $B_{n+1} = min(B_n, A_n)$, $C_{n+1} = C_n$ ***Continue to step n + 1.*** Calculate gcd(33, 32) with steins algorithm. (provide stepwise answer)	10 10	CO1-3

Theory-ISE-Quiz



Lab-ISE-Rubrics

Performance Indicator (PI)	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Preparedness [Marks = 3]	Has referred textbook as well as some reference books and good theoretical background knowledge and able to solve any problem. **(Marks= 3)**	Has knowledge of few textbooks and referred some of them and able to solve some problems theoretically based on given experiment. **(Marks= 1 – 2)**	Does not know textbooks and what the aim of experiment is. **(Marks= 0)**
Experimental Procedure [Marks = 3]	Able to make any changes, as and when asked to. **(Marks= 3)**	Able to make few changes. **(Marks= 1 – 2)**	Not able to make any changes. **(Marks= 0)**
Result [Marks = 4]	Application functioning is as expected in the aim. All the requirements are fulfilled. **(Marks= 3 - 4)**	Application functioning is as per the aim but some of the requirements are not fulfilled. Partially running the experiment with some limitations. **(Marks=1-2)**	Application functioning is not as expected in the aim. **(Marks= 0)**

C8. Quality of assessment tools and extent of attainment of learning outcomes

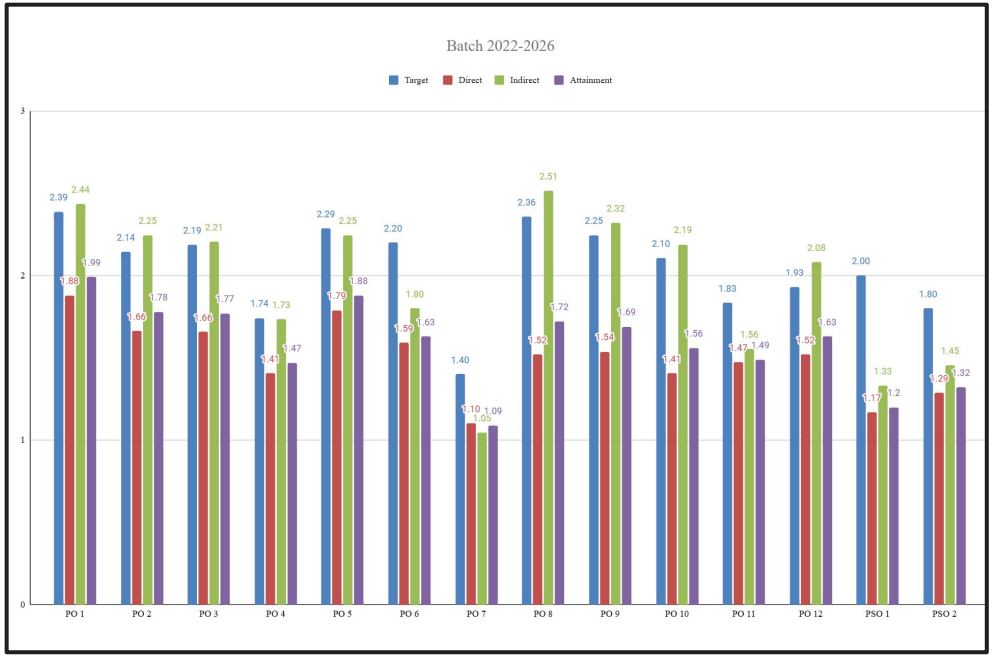
BATCH 2021-25

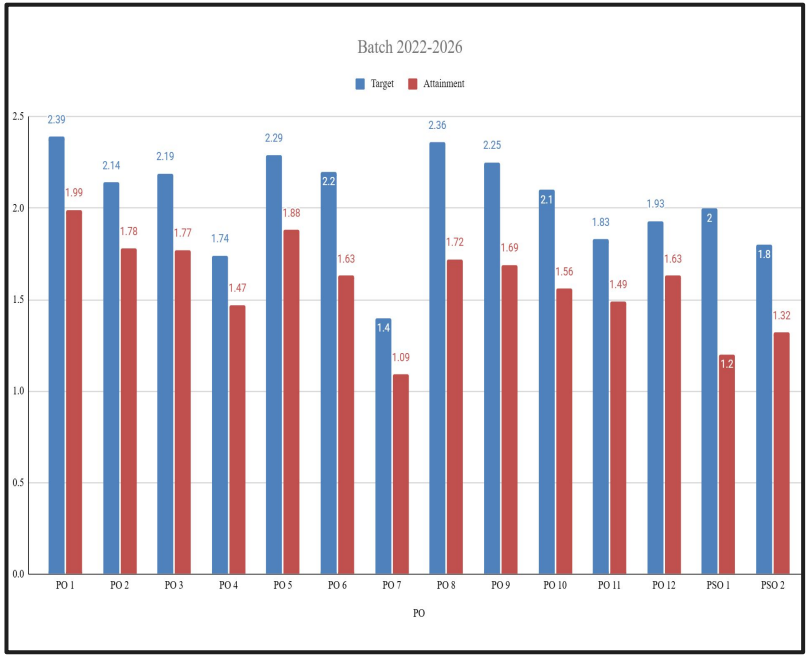
PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
Target	2.29	2.21	2.33	1.85	2.45	1.86	1.13	2.07	2.02	2.13	1.83	1.82	2.15	1.72
Direct	1.75	1.76	1.73	1.67	1.90	1.49	1.35	1.22	1.59	1.69	1.79	1.49	1.39	1.16
Indirect	2.40	2.48	2.37	2.22	2.41	1.76	1.06	2.23	2.33	2.58	2.37	2.44	1.87	2.26
Attainment	1.88	1.91	1.86	1.78	2	1.54	1.29	1.42	1.74	1.87	1.9	1.68	1.49	1.38

BATCH 2022-26

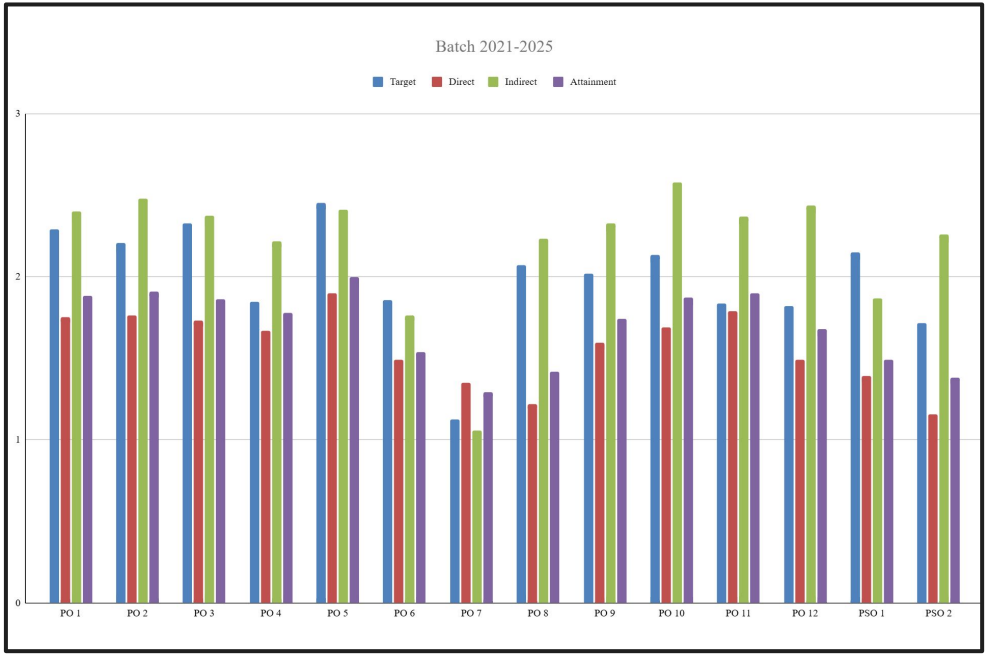
PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
Target	2.39	2.14	2.19	1.74	2.29	2.20	1.40	2.36	2.25	2.10	1.83	1.93	2.00	1.80
Direct	1.88	1.66	1.66	1.41	1.79	1.59	1.10	1.52	1.54	1.41	1.47	1.52	1.17	1.29
Indirect	2.44	2.25	2.21	1.73	2.25	1.80	1.05	2.51	2.32	2.19	1.56	2.08	1.33	1.45
Attainment	1.99	1.78	1.77	1.47	1.88	1.63	1.09	1.72	1.69	1.56	1.49	1.63	1.2	1.32

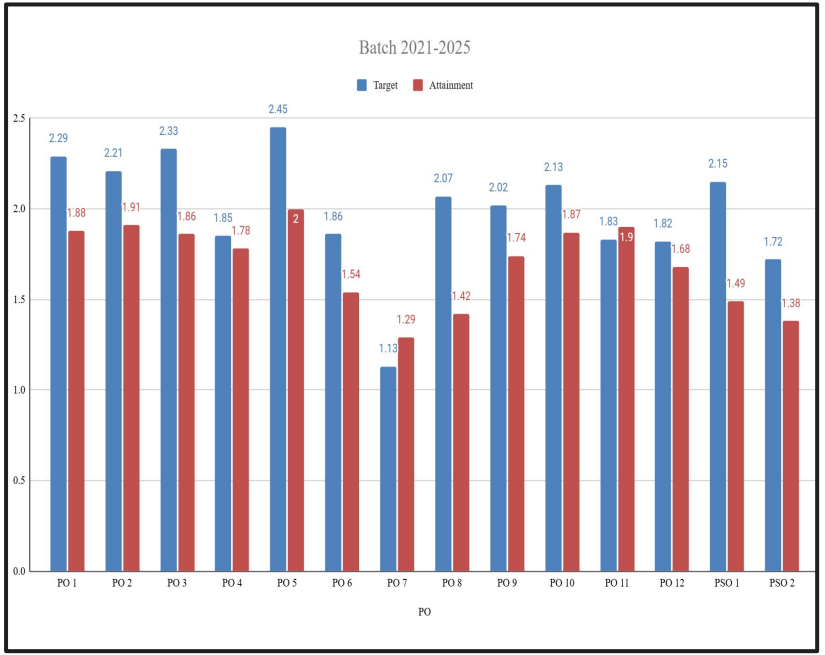
C9. Department checks the course attainment every semester, PO/PSO attainment every year, PEO attainment at an interval of 4 years.





C9. Department checks the course attainment every semester, PO/PSO attainment every year, PEO attainment at an interval of 4 years.





CRITERIA D

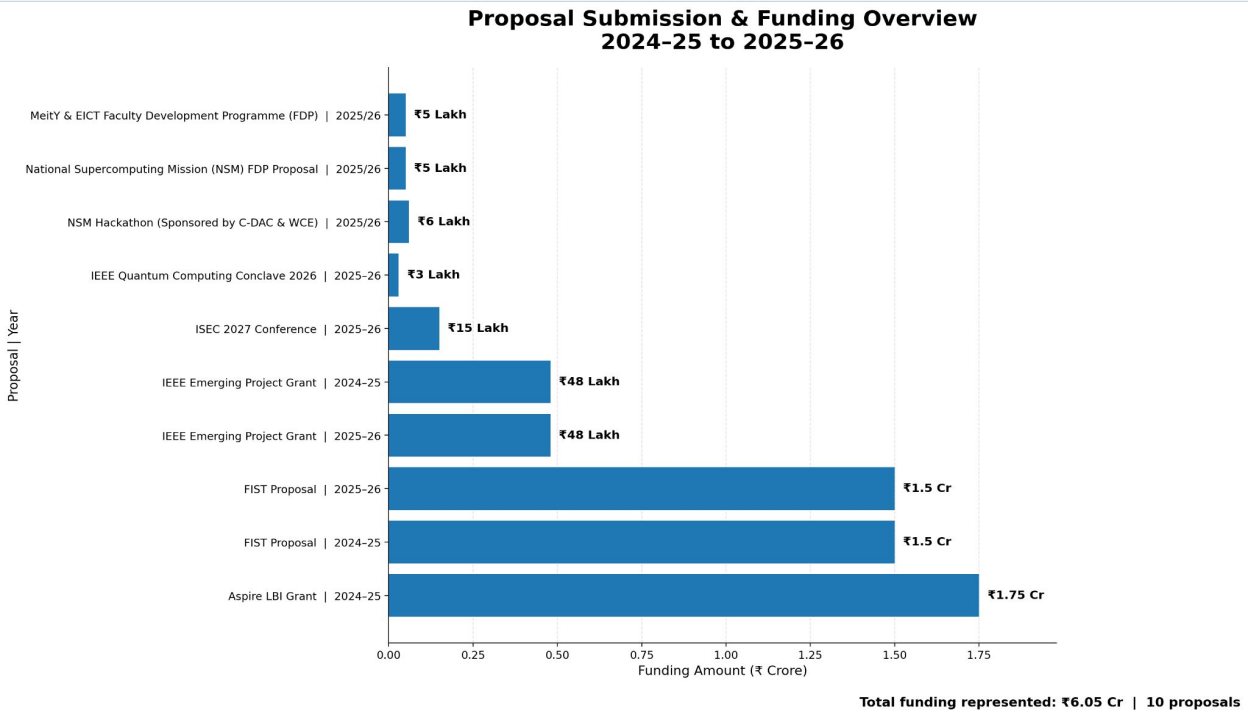
RESEARCH & CONSULTANCY

Faculty publications, sponsored research, consultancy, student IPR

D.Research

1. Teachers are encouraged to participate in the FDP, Workshop, STTP.
2. Every Faculty publishes paper in reputed Conference or Journal.
3. Department organizes workshop, FDP for upgradation.
4. Research related activities : Students are encouraged to be a part of Technical Research Paper Competitions.
5. Students are encouraged to work on Research Based Mini Projects.

D.1. Research related activities of faculty. Writing proposals for sponsored research projects, research focused projects for UG/PG students, PhD program



D.1. Research related activities of faculty. Writing proposals for sponsored research projects, research focused projects for UG/PG students, PhD program

- Students are encouraged to apply for research internships at reputed institutes and research organizations such as IITs, BARC, ISRO, NTRO, etc., providing them with exposure to practical, real-world research problems and industry/research practices.

Sr No	UID	Student Name	Project Title	Year	Company	Mentor
1	2023300139	Raj Kalpesh Mathuria	Proprietary File System Reverse Engineering and Data Extraction Framework	2025-26	NTRO	Sameera Mishra
2	2023300195	Vishwajit Sambhaji Sarnobat	Proprietary File System Reverse Engineering and Data Extraction Framework	2025-26	NTRO	Sameera Mishra

D.2. Industrial projects, Sponsored research, collaborations and consultancy

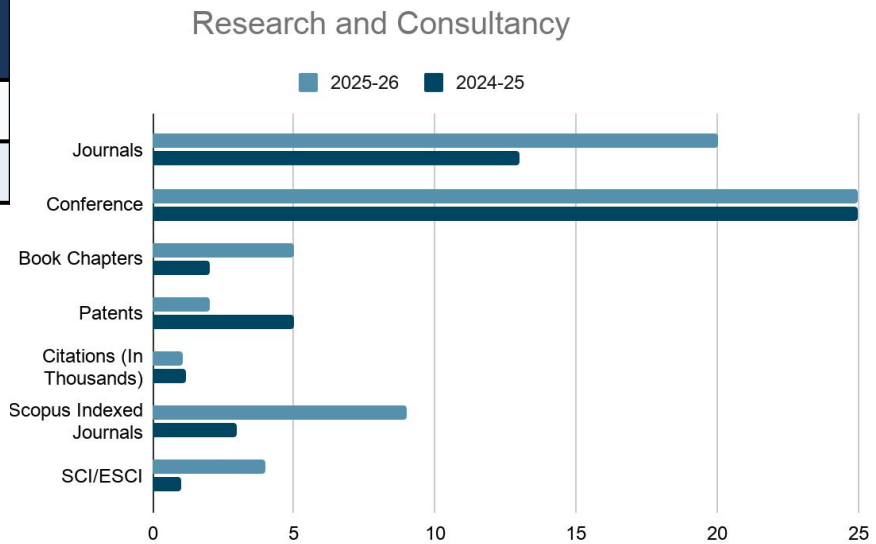
Consultancy Projects and Revenue Generated

Sr. No.	Faculty Name	Title of Project	Year	Revenue Generated
1	Dr. Anant V. Nimkar	lOT GATEWAY DESIGN	2024 - 25	Rs. 10.2 Lakhs

D.3. Faculty publications, quality, citations, h-index, and IPR, books and book chapters, monograms.

Year	Journals	Conference	Book Chapters	Patents	Citations
2025-26	20	25	5	2	1077
2024-25	13	25	2	5	1168

Year	Scopus Indexed Journals	SCI/ESCI
2025-26	09	04
2024-25	03	01



D.4. Faculty achievements, awards and recognition

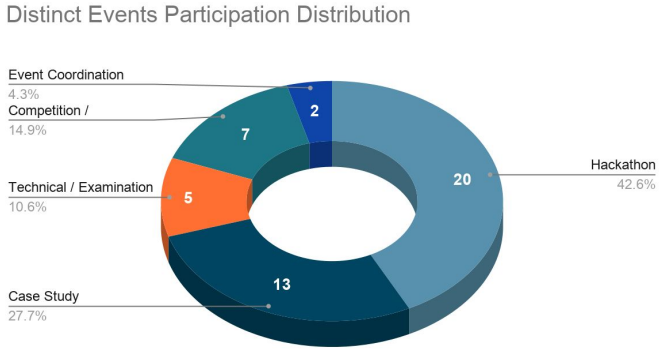
Name of Faculty	Achievement
Prof. Anand Godbole	NPTEL Discipline Star Award
Dr. Pramod Bide	Best Innovative Teaching Award by L&T Edutech
Dr. Jignesh Sisodia	Best Software Project Award by Cantilever
Prof. Govind Gaundalkar	Best Question Paper Award for the subject Digital Systems
Dr. Jignesh Sisodia, Prof. Abhijit Salunke	Best Question Paper Award for the subject Computer Communication and Networks

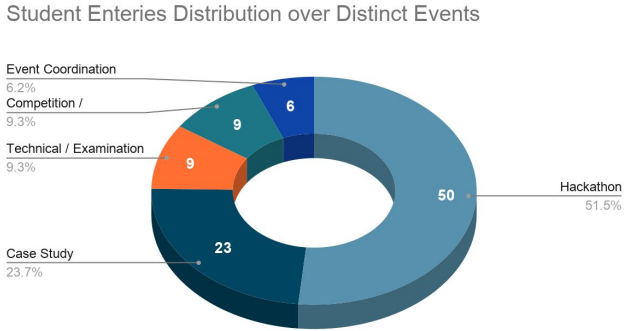
D.5.Student publications and IPR

Students	2024-25		2025-26	
	Journals	Conferences	Journals	Conferences
UG	9	18	10	1
PG	1	2	2	29
Phd	2	2	2	0

D.6. Student participation and success in the technical competitions of repute, AICTE hackathon, ROBOCON, etc.

Competition Category	Distinct Events	Student Entries
Hackathon	20	50
Case Study Competition	13	23
Technical / Examination	5	9
Competition / Recognition	7	9
Event Coordination	2	6





CRITERIA E

STUDENT SUPPORT & PROGRESSION

Mentorship, career counseling, student clubs, competitive exam success

E.1.Department has well planned mentorship scheme and every student is reached. There is observed periodicity. The problems/concerns are immediately addressed, and this process is well documented.

Structured Mentorship

- Department-wide mentorship mechanism covering all students
- Faculty mentors provide academic guidance and student support
- Student concerns identified through mentoring interactions, attendance and academic monitoring

Evidence Maintained

- Student Attendance & Academic Records
- Concern / Issue Register
- Action Taken Reports

Periodic Monitoring & Support

- Student attendance and academic performance are reviewed
- Concerns are recorded and appropriate interventions and referrals are initiated based on the nature of the concern
- Follow-up and action taken are documented

Continuous Improvement

- Departmental review of student concerns and interventions
- Consolidation of mentorship records strengthened for systematic audit documentation
- Standardized student-wise interaction and follow-up records proposed for future cycles

E.2.Department has effective mechanisms for academic monitoring both formally and informally. The feedback, suggestions are considered duly.

- The department regularly monitors academic activities through formal and informal mechanisms.
- Feedback and suggestions from students and faculty are reviewed and considered for improvement.
- Class Representatives (CRs) help in identifying and communicating academic issues and concerns.
- Course Exit Feedback is collected by the respective faculty at the end of the course to understand students' learning experience and identify areas that can be improved.
- Faculty feedback is collected after the Mid-Semester and End-Semester examinations.
- The feedback is analyzed, and the observations are shared with the concerned faculty to identify areas for further improvement in teaching and academic delivery.

E.3.Department has a 360-degree feedback system, student feedback on academics, admin etc, grievances/ redressal and its effectiveness

DAPC

- It is committee responsible Academic Planning including and Decide SA values for Examination.

Academic Monitoring Committee

- **Objective:** Ensure effective teaching-learning, quality laboratory practices, timely resolution of concerns, and continuous academic improvement.
- **Committee Members:** three members and one coordinators
- **Key Responsibilities:** Conduct random inspections of theory classes and laboratories., Monitor punctuality, preparedness, teaching effectiveness, discipline and timetable adherence. Verify laboratory support, equipment, software, safety and Technical Assistant responsibilities. Identify and document academic/laboratory issues and student concerns., Recommend corrective and preventive actions for improvement.
- **Reporting**: Fortnightly consolidated report by the Coordinator to the HoD; major issues reported immediately.

E.4.Department maintains close relations with the alumni and derives best possible benefits with their association. For example, co teaching, special lectures, co supervision of projects etc.

- The Alumni are involved in evaluation of Internship Students undergoing Internship at various Industries. in year 2024-25 , 18 Alumni evaluates the students. in year 2025- 26 , 11 alumni participated in evaluation.
- Dr. Sudhir Dhage, have arranged the Almni Talk on “Exploring and Preparing for management and consulting careers” which was delivered by Soumyadeb Bhattacharjee- Strategy and Management Consultant.

Year 2025-2026

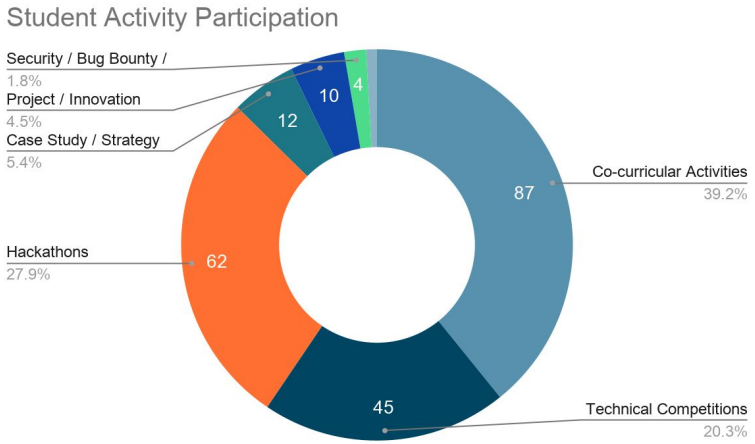
SN	Faculty Name	Subject	Class	Date	Mode
1	Mrs. Kurhade Swapnali	Smart Industries Powered by AI: Case Studies and Applications	BE Comps	30/10/2025	Online
2	Prof. Anant Nimkar	Health Care Balancing Innovations and Responsibility	BE Comps	30/11/2025	Online
3	Prof. Nataasha Raul Prof, Anand Godbole	Tries: Structures Operations and its application	SE Comps	21/10/2025	Online
4	Dr. Sudhir Dhage	Next Generation AI Seminar	TE comps	16/03/2026	Offline
5	Prof. Soni Bhamber	Python DB programming	Mtech Comps	15/12/2025	Online
6	Mrs. Ramteke Jyoti Yashwant	AI and Gen AI: Trends and Future in Industry	TE comps	22/11/2025	Online
7	Mrs. Ramteke Jyoti Yashwant	Recent Trends in HPC and Future Quantum Technologies	BE Comps Mtech Comps	19/04/2026	Online

Year 2024-2025

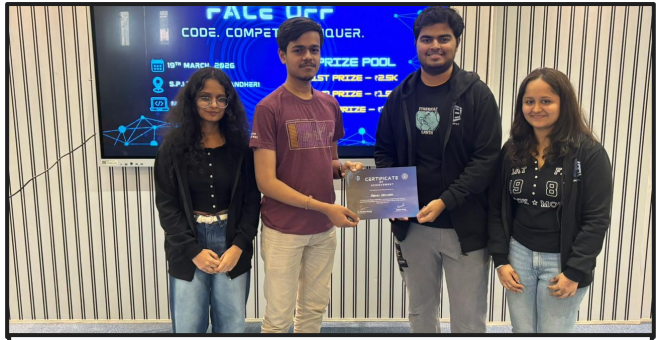
SN	Faculty Name	Subject	Class	Date
1	Prof. Anant Nimkar	IoT Security	BE Comps	05/10/24
2	Prof. Anant Nimkar	Cyber Security	BE Comps	22/10/24

E.5.Students' participation in extracurricular and cocurricular activities, technical and cultural events, quizzes, workshops, seminars, conferences, exhibitions, tech competitions like ROBOCON, BAHA etc.

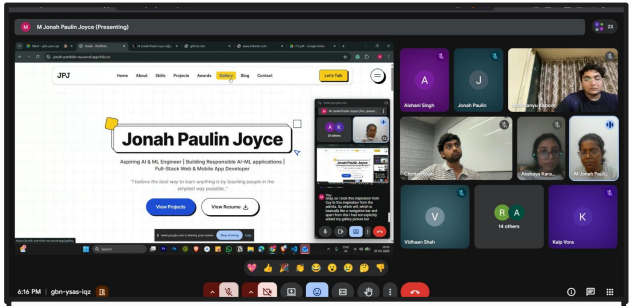
Sr. No.	Activity Category	No. of Students
1	Co-curricular Activities	**87**
2	Technical Competitions / Tech Events	**45**
3	Hackathons	**62**
4	Case Study / Strategy Competitions	**12**
5	Project / Innovation Competitions	**10**
6	Security / Bug Bounty / CTF	**4**
7	Academic / Competitive Examinations	**2**



E.6.Institute promotes the culture of student clubs. Students are motivated and encouraged to start dedicated, focused student clubs with diversified objectives.- FACE - Forum for Aspiring Computer Engineers



FaceOff



Codefolio



Donation Drive 1



Donation Drive 2

E.6.Institute promotes the culture of student clubs. Students are motivated and encouraged to start dedicated, focused student clubs with diversified objectives.- CSI Events & Technical Workshops (2024–25)

Name of Event	Date
Tech Roadmap	30th September 2024
GSoC Session	1st October 2024
Git GitHub Session	1st October 2024
Source Start	1st - 31st October 2024
CodeHousie	17th October 2024
Minithon	20th October 2024
Mini Source Start	23rd October 2024
Frontend Odyssey	1st February 2025
SQL and API Workshop	4th February 2025
rAPId	5th February 2025
UnSQLved	6th February 2025
Main Speaker Session	7th February 2025
S.P.I.T Hackathon	8th-9th February 2025
Code Uncode	20th February 2025
SE Hackathon	19th-20th April 2025



SPIT Hackathon 2024-2025



GSoC session 2024-2025

E.6.Institute promotes the culture of student clubs. Students are motivated and encouraged to start dedicated, focused student clubs with diversified objectives.- CSI Events & Technical Workshops (2025–26)

Name of Event	Date
ML Sprint	17th February 2026
CTRL Fix+ Takedown	18th February 2026
Design Disaster	19th February 2026
UnSQLved	20th February 2026
SPIT Hackathon	21st February 2026
SE Hackathon	18th April 2026
Git and GitHub Workshop	1st October 2025
Code Housie	16th October 2025
SomETHing Decentralised	3rd November 2025
Pixel Perfect	14th February 2026
Code it Ralph	16th February 2026



GitHub Session 2025 - 2026



SPIT Hackathon 2025 - 2026

E.6.Institute promotes the culture of student clubs. Students are motivated and encouraged to start dedicated, focused student clubs with diversified objectives.- IEEE CS-SPIT

IEEE CS-SPIT





Brain Drain

On 17th November 2025, IEEE Computer Society S.P.I.T. organized Brain Drain, an ice-breaker event for the latest first years. The event was conducted to introduce first-year students to the IEEE C.S community and encourage participation through interactive activities.





Research Ideation

On January 28, 2026, IEEE Computer Society SPIT hosted "How to Define a Research Problem" in Room 008, drawing around 80 participants. Speakers Dr. Fitri Yakub (IEEE Malaysia/MJIIT, joining virtually), Dr. Sudhir Dhage, and Dr. Pramod Bide introduced undergraduate students to research fundamentals and academic writing to encourage research-oriented projects.

E.6.Institute promotes the culture of student clubs. Students are motivated and encouraged to start dedicated, focused student clubs with diversified objectives.- IEEE CS-SPIT

IEEE CS-SPIT









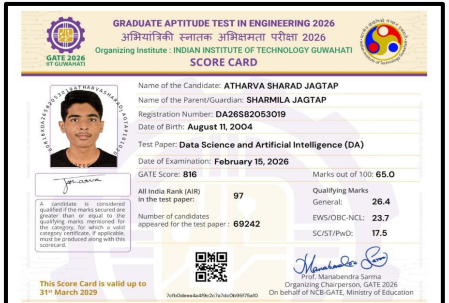
Quantum Computing Conclave

The Quantum Computing Conclave organized on 20-21 February 2026 successfully brought together renowned experts, researchers, faculty members, and students to foster knowledge exchange and collaboration in the field of quantum technologies.

AIRAVAT 3.0

SPIT AIRAVAT 3.0, the third edition of IEEE Computer Society SPIT's flagship AI hackathon, was conducted on April 11-12, 2026 at Sardar Patel Institute of Technology, Mumbai. This 24-hour national-level hackathon brought together the brightest young innovators to solve real-world problems using emerging technologies.

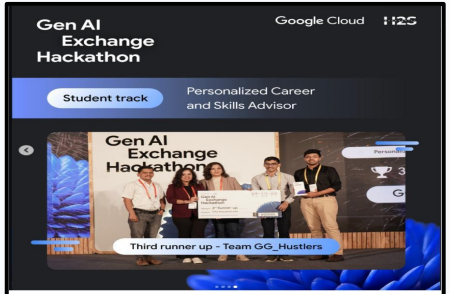
E.7.Student achievements, awards and recognitions.


GRADUATE APTITUDE TEST IN ENGINEERING 2026
अभियांत्रिकी स्नातक अभिक्षमता परीक्षा 2026
Organizing Institute : INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI
SCORE CARD

Name of the Candidate: ATHARVA SHARAD JAGTAP
Name of the Parent/Guardian: SHARMILA JAGTAP
Registration Number: DA26S82053019
Date of Birth: August 11, 2004
Test Paper: Data Science and Artificial Intelligence (DA)
Date of Examination: February 15, 2026
GATE Score: 816 Marks out of 100: 65.0
All India Rank (AIR) in the test paper: 97
Number of candidates appeared for the test paper: 69242
Qualifying Marks
General: 26.4
EWS/OBC-NCL: 23.7
SC/ST/PwD: 17.5
This Score Card is valid up to 31st March 2029
Prof. Manabendra Sarma
Organizing Chairperson, GATE 2026
On behalf of NCB-GATE, Ministry of Education

AIR-97 in GATE DA, with a Score of 816 IIT Guwahati


BARCLAYS HACK O HIRE WINNER Gera Commerzone SEZ, Building G2, 5th to 12th Floor (Part), Survey No. 65, Kharadi, Pune, Maharashtra 411014



GenAi Exchange Hackathon Third Runner Up Google Cloud


(1st position) at the hackathon hosted by BITS Pilani, Rajasthan

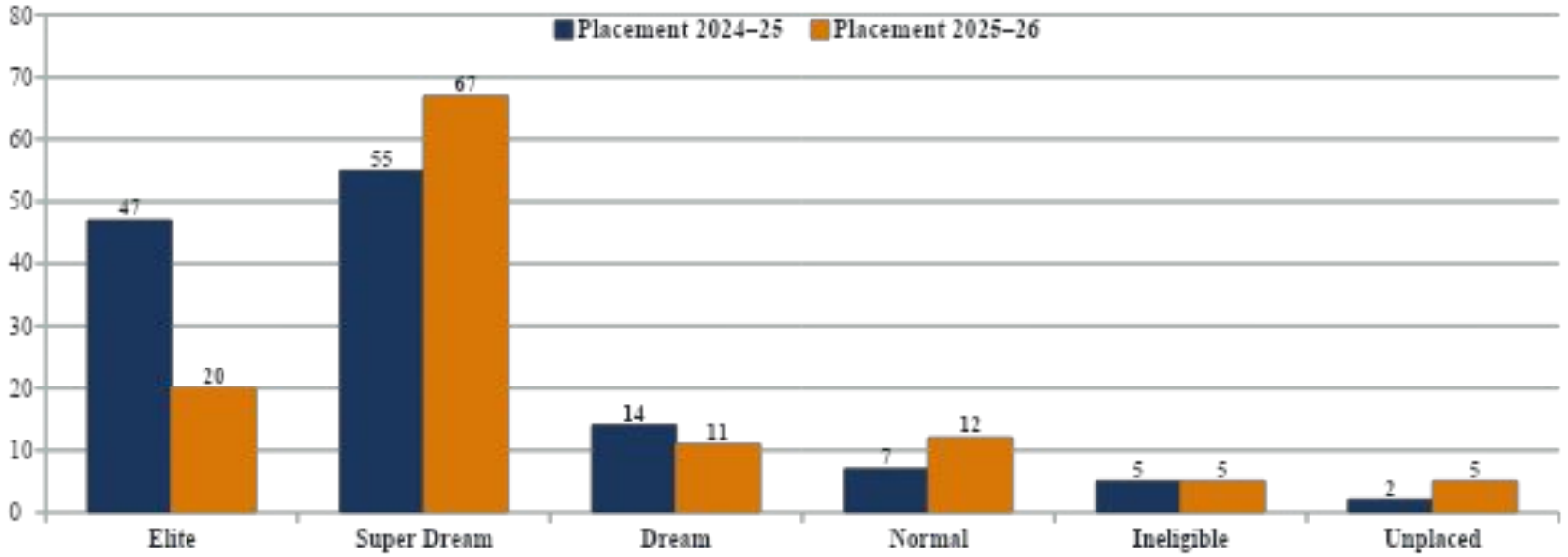


National Winner – EY Techathon Ernst & Young


Adappt 4.0 Hackathon Winner – 1st Place NMIMS

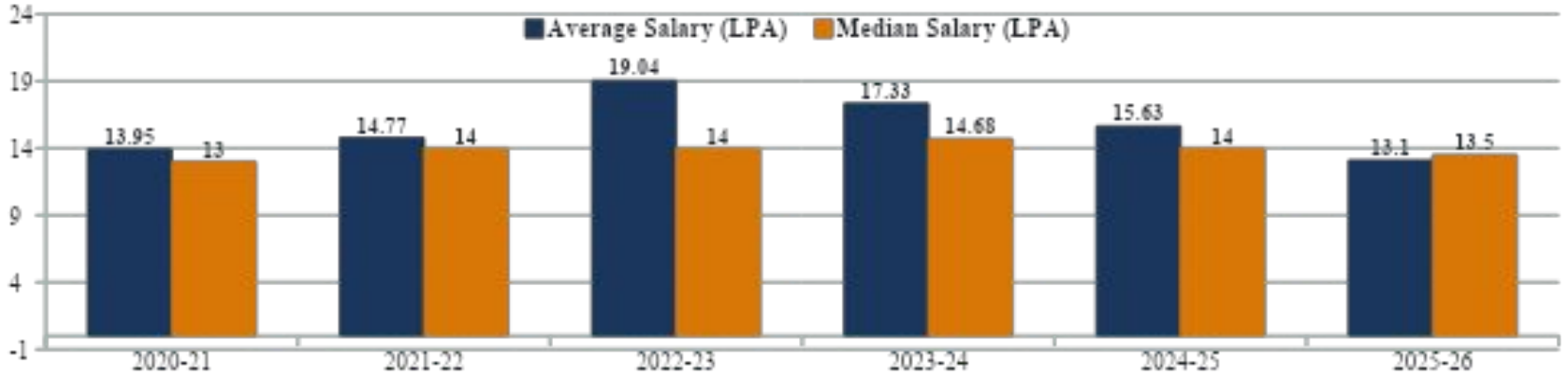
E.8. Student Competency: Placement Statistics & Salary Tiers

ELITE	SUPER DREAM	DREAM	NORMAL
≥ 18 LPA	9 to 17 LPA	6 to 9 LPA	< 6 LPA



Placement Salary Trends: Highest, Average & Median (2020–26)

Metric (LPA)	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26
Highest Salary	42	30.5	110	50	50	33
Average Salary	13.95	14.77	19.04	17.33	15.63	13.1
Median Salary	13	14	14	14.68	14	13.5
Lowest Salary	4	6.3	5	3	3.5	3



E.9. Department takes significant efforts to fetch industry sponsorship for UG/PG projects and ensures the quality execution of these projects.

- The department encourages students to take up research internships and project work with reputed organizations such as IITs, ISRO, BARC, and NTRO.
- These opportunities give students valuable practical exposure and help them gain experience in working on real-world problems.

CRITERIA F

GOVERNANCE & ADMINISTRATION

Institutional governance, faculty development, quality assurance

F.1. Department has well thought Vision-Mission and has carefully articulated Program Education Objectives and Program specific objectives for all the programs run by the department.

Vision

"To impart the education that graduates globally competent, socially sensitive, contributory computer professionals."

Mission

- To provide rigorous, conducive academic environment and constantly innovate in the design of the curriculum and its deployment (T-L-E-A Processes).
- To collaborate with industries and academia to strengthen education and ensure relevant research.
- To focus on experiential learning and inculcate problem solving, leadership, professional ethics, teamwork and sensitivity to the real-world problems.
- To ensure a professional growth mindset that translates to lifelong learning for all.

F.1. B.Tech.: Program Educational Objectives (PEOs) & Program Specific Outcomes (PSOs)

B.Tech. — Program Educational Objectives (PEOs)
PEO1: Graduates will demonstrate proficiency in computing technologies and innovate in design, development, deployment, and maintenance of serving various sectors of economy.
PEO2: Graduates will be effective communicators, inspiring leaders, aspiring team members, and trailblazers of computational technologies.
PEO3: Graduates will be contributory professionals demonstrating lifelong learning attitude and sensitivity to society problems.

B.Tech. — Program Specific Outcomes (PSOs)
PSO1: Possess strong foundation of Computer Engineering (Algorithms, Data Structures, AIML, Databases, Networks, Security etc.) and apply the same for finding efficient and feasible solutions for real-life problems.
PSO2: Be motivated to engage in lifelong learning, adopt strong values, and develop skills in project and finance management. Showcase leadership, teamwork, and social awareness, while becoming proficient in databases, application software, computer networks and security, cloud services, storage, data manipulation, etc.

F.1. M.Tech.: Program Outcomes (POs) & Program Educational Objectives (PEOs)

M.Tech. — Program Outcomes (POs)	M.Tech. — Program Educational Objectives (PEOs)
PO1: An ability to independently carry out research / investigation and development work to solve practical problems.	PEO1: Graduates will demonstrate technical competency and maintain software products, services and systems with strong analytical and problem-solving skills.
PO2: An ability to write and present a substantial technical report/document.	PEO2: Graduates will be adept in collaborating on interdisciplinary projects fostering innovations.
PO3: Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.	PEO3: Graduates will exhibit effective leadership, teamwork, and lifelong learning.

F.2. Department has a strategic plan which is broadly in line with the institute's strategic plan.

- Establish minimum two advanced laboratories with modern hardware/software, AI, IoT, Cloud, Cybersecurity and GPU resources.
- Enable at least 15% of graduating students to pursue higher studies through mentoring, GRE/GATE guidance and university awareness.
- Ensure 90% faculty possess Ph.D. qualification or are in advanced stages of completion.
- Achieve at least two quality journal publications every year in reputed indexed journals.
- Conduct minimum five expert talks annually on emerging technologies.
- Submit at least two proposals for FDP/STTP/Workshop/Research funding each year.
- Every faculty member enrolls in at least one NPTEL course every semester.

F.3. Department has a documented manual for all processes, procedures and information is disseminated to all stakeholders. Transparency is maintained to the best possible level.

Department Process Documentation & Transparency :

Institute-Level Framework

- Major institutional processes are defined, documented and governed at Institute level through the SPIT Process Manual.
- The Department follows these approved institutional processes and guidelines wherever applicable.
- S.P.I.T. Process Manual - https://drive.google.com/file/d/1HcyGdIs4PE4ojHEn-NNUE-DZo91bLj-D

Department-Level Academic Processes:

The Department maintains and implements documented academic processes covering:

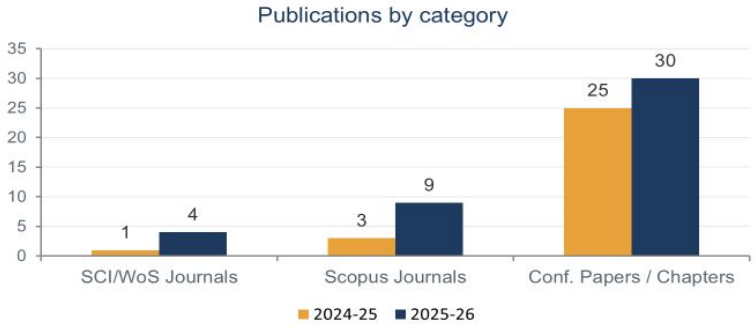
- Curriculum & syllabus implementation
- Teaching–Learning and Academic Planning
- Internal assessment, examinations & evaluation
- Laboratory and project evaluation
- CO/PO/PSO attainment
- Academic audit, feedback and continuous improvement

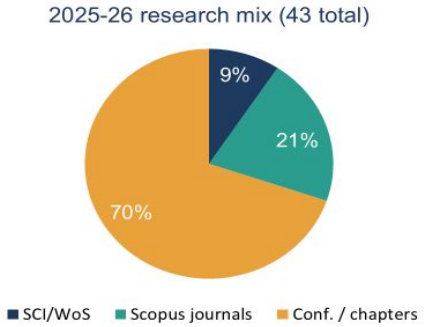
F.4. Grievances are duly addressed for students, faculty, and staff.

- All the Grievances of the students are duly addressed on regular basis during the meetings with the class representatives and Batch representatives of the respective classes
- All the exam related grievance are solved during the open house and are resolved with the subject teachers.

F.5. Department Meets the Target Set by the Institute

Research & Industry Interaction: Strong Upward Trend

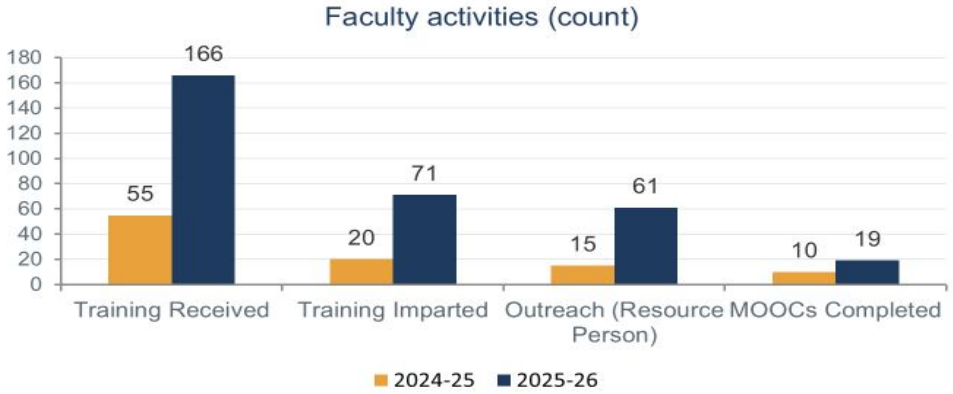


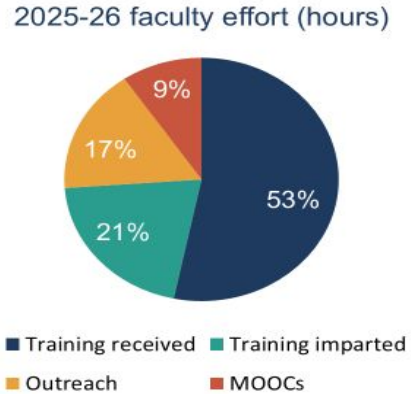


Research output rose 48% in one year, with the biggest jump in quality journals (SCI/WoS 1 to 4, Scopus 3 to 9). Industry expert lectures more than tripled, strengthening industry-academia links.

29 → 43	1 → 4	2 → 7
Total research output (+48%)	SCI/WoS journals (4x)	Industry expert lectures (+250%)

F.5. Faculty Development: Sharp Growth Across the Board





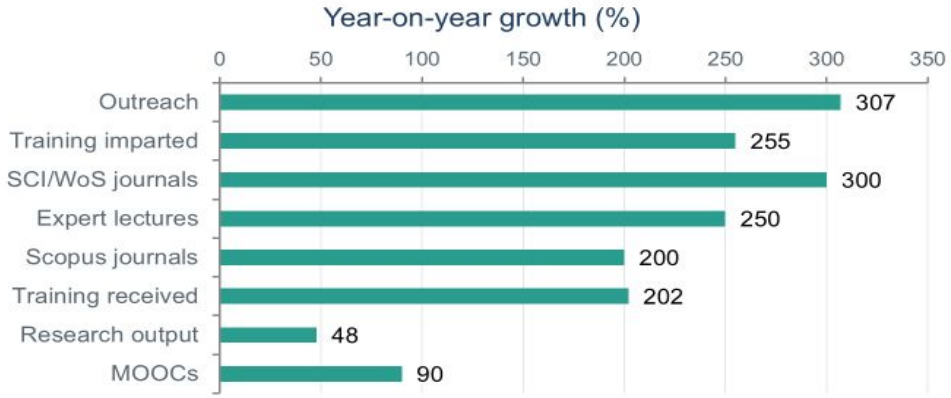
Every faculty development indicator grew, led by outreach as resource person (4x) and training imparted (3.5x). Faculty invested over 8,600 hours in learning, teaching and outreach in 2025-26.

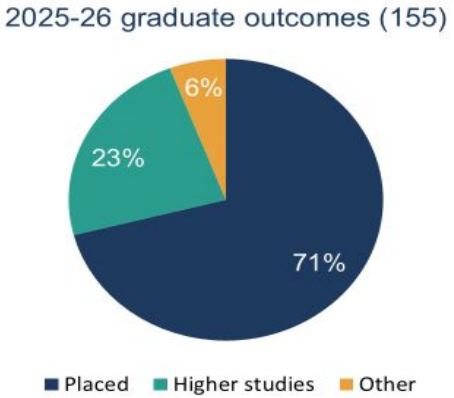
+202%
Training received (55 → 166)

+255%
Training imparted (20 → 71)

+307%
Outreach (15 → 61)

F.5. Overall Growth Snapshot & Student Outcomes





Growth is broad-based: seven of eight indicators grew 90% or more. Higher-studies enrolment nearly tripled (13 to 36), crossing the 15% objective; 71% of graduates were placed.

13 → 36
Higher studies (8.5% → 23%)

110 / 155
Graduates placed in 2025-26 (71%)

7 of 8
Indicators grew 90%+

F.6. Department values continuous learning, value addition and takes sincere efforts in that direction. July 2025 - June 2026

	Count	No. Of Hours	No. Of Days
Training Received	166	4621	1235
Training Imparted	71	1788	155
Outreach As Resource	61	1442	90
MOOC Completed	19	825	828



F.6. Department values continuous learning, value addition and takes sincere efforts in that direction. July 2024 - June 2025

	Count	No. Of Hours	No. Of Days
Training Received	55	463	67
Training Imparted	20	38	17
Outreach As Resource	15	20	13
MOOC Completed	10	48	6



NVIDIA

Certificate of Competency

This certificate is awarded to

Reeta Koshy

for demonstrating competence in the completion of

Fundamentals of Accelerated Computing with CUDA Python

Greg Estes
Vice President, NVIDIA

ATAL/2025/1736835553

Skill India | ATAL | AICTE

ALL INDIA COUNCIL FOR TECHNICAL EDUCATION

Nelson Mandela Marg, Vasant Kunj, New Delhi -110070

AICTE Training and Learning (ATAL) Academy

Certificate

It is certified that ***Dr. Kailas K Devadkar, Faculty members of the AICTE approved institutions*** *of* ***SPIT Andheri*** *has successfully participated & completed AICTE Training And Learning (ATAL) Academy Faculty Development Program on* ***The Role of Artificial Intelligence in Sustainable Waste Management.*** *at* ***FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING*** *from* ***10/02/2025*** *to* ***15/02/2025.***

Prachi Harshal Dalvi
Assistant Professor Level (AICTE Institute), Coordinator
FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Dr. Sunil Luthra
Director & Bureau Head
Training and Learning Bureau, AICTE

F.7.Department ensures that the inputs/feedback received from various procedures (internal and external audits, accreditation visits, QP analysis) are analyzed, discussed, deliberated, and implemented to the best possible level as a part of continuous improvement.

- **Department meeting** is conducted after every audit to discuss and analyze all the inputs/feedbacks received from various committees like internal audit, external audit NBA and NAAC.
- **Question Paper analysis** is maintained for each course by the respective course coordinators

Process - Question Paper analysis

- The process of audit of Question paper is carried out by the examination section. and then feedback is given to concerned teacher of the subject.
- H.O.D formally gives instruction to concerned teacher to improve upon the drawn question paper.
- Faculty is asked to update the question paper and asked to follow the guidelines of Bloom's taxonomy , CO-PO mapping.

CRITERIA G

SWOC ANALYSIS & STRATEGIC GOALS

Strengths, Weaknesses, Opportunities, Challenges and Future Roadmap

SWOC Analysis

S | STRENGTHS

Established Academic Ecosystem

- NEP 2020-aligned, outcome-based curriculum with flexible electives and experiential learning.
- Active research, IPR, projects and higher-studies pathways.
- Strong industry and alumni engagement in learning and evaluation.

W | AREAS TO STRENGTHEN

Scale & Systematize

- Deepen sustained sponsored research, consultancy and external project funding.
- Further strengthen systematic documentation of mentoring and continuous-improvement actions.
- Broaden academic and research engagement.

O | OPPORTUNITIES

Emerging & Collaborative Domains

- Scale learning and research in AI/ML, Generative AI, Cloud, Cybersecurity and Quantum Computing.
- Expand industry, research-institute and externally sponsored project collaborations.
- Strengthen innovation, hackathons, research publication and entrepreneurship pathways.

C | CHALLENGES

Rapidly Changing Ecosystem

- Keep curriculum, laboratories and faculty capabilities aligned with fast technology cycles.
- Scale experiential learning and mentoring with increasing student intake.
- Compete effectively for research funding, talent and industry partnerships.

Goals of the Department

01

Research & Sponsored Projects

Expand funded research, consultancy, IPR and interdisciplinary project activity.

02

Industry–Academia Integration

Deepen industry co-teaching, sponsored projects, internships and expert engagement.

03

Emerging Technology Readiness

Continuously strengthen learning and infrastructure for AI/ML, GenAI, Quantum, Cloud and Cybersecurity.

04

Innovation & Entrepreneurship

Convert projects and hackathons into prototypes, publications, IPR and innovation pathways.

05

Global & Higher-Study Engagement

Expand research internships, academic collaborations and pathways to reputed higher-study institutions.

06

Continuous Improvement

Use audit, feedback and outcome evidence to strengthen mentoring, laboratories and academic processes.

Strengths

01

Academic Excellence

Autonomous, NEP 2020-aligned curriculum with OBE, flexible electives, research and industry internships.

02

Faculty Excellence

Experienced faculty with active participation in research, publications, IPR, FDPs and professional development.

03

Research & Innovation

2025–26: 20 journal papers, 25 conference papers, 5 book chapters and 2 patents.

04

Industry & Alumni Connect

Industry/alumni participation through co-teaching, internship evaluation, expert sessions and skill initiatives.

05

Student Outcomes

2025–26 placement rate of 98.21%; higher-studies participation increased to 25 students.

06

Learning & Infrastructure

Upgraded laboratory infrastructure, digital learning resources, project-based learning and structured student support.



THANK YOU!

Department of Computer Engineering

Sardar Patel Institute of Technology, Mumbai